Company Confidential

Oct 26, '60

Memo for File

Subject: What should FSD's Strategy be [redacted] regarding STRETCH and Super-STRETCH? -- First Pass --

I. Background.

The STRETCH program has [redacted] suffered a great many ups and downs [redacted] since it was started in 1955. The Corporation has reorganized several times since then and the STRETCH program has not always enjoyed uniform support. The program grew out of a competetive situation at the AEC's Livermore Lab, in which they invited bids on a 2 Mc computer. [redacted] The contract was won by Sperry-Rand and [redacted] eventually produced the LARC. IBM [redacted] later obtained a contract from AEC Los Alamos for a 10 Mc computer. The contract [redacted] for a fixed fee of $4.3 million was signed in Dec 1956, at that time IBM realized that the program would cost considerably more than this -- but no one guessed how much more -- [redacted] at the same time negotiations were underway with Bu Ships for another contract for a special purpose STRETCH-like machine later called HARVEST. This program got underway during 1957 [redacted] and proceeded in parallel with STRETCH using the same circuits & many of the same planners.

Unfortunately it is hard to imagine two requirements more different than those of AEC Los Alamos and the HARVEST customer. The repeated attempts to pull these [redacted] segregate together into one logical design ~~caused a fair, in my opinion, irreparable damage to the STRETCH~~ was an almost impossible task, full of many compromises.

one result of this "shotgun marriage" has been that each of the customers now feels that he has been put in second place to the other. Incidentally, the STRETCH system, without Harvest, runs about two degrees of the to 200,000 transistors.

Because it was breaking trail in technology and logical design and because it was costing the Corporation so much IBM money, it was easy to criticize the STRETCH program at all levels -- and it was. The fact that the 7090 sprang almost full grown from STRETCH components has been widely overlooked.

Last winter it was realized that STRETCH would not meet its delivery dates of June 1, 1960 for STRETCH and Sept 15, 60 for HARVEST. a crisis developed resulting in ~~present~~ increasing the rate of expenditure and amount of overtime. At the present time about as many people as can be put on the project are working on it. In spite of this, the STRETCH program has slipped in time and performance.

II, Present Status of STRETCH

Despite these difficulties within the Company, STRETCH has enjoyed a tremendous popularity in the outside world. It is considered the unrivaled master of the computing field. In spite of a very high price placed upon it by DSD, two additional systems have been sold and several more are in final negotiations. There is no question but that the IBM reputation is a large factor in this acceptance,

The present delivery plans are still uncertain. The AEC Los Alamos machine could have been delivered about Nov 1, 1960 without its disk file. Los Alamos refused to accept it until the File is installed (— extended to be about 1 Feb 1961.) and their acceptance Tests have been run. The Harvest is in more uncertain status scheduleswise. A new date is to be furnished the customer by DSD on Nov 15, 1960.

III. Present plans for STRETCH

Subsequent STRETCH systems after the two models are to be built at the Kingston plant. The first two are now under construction. Approximately 100 Kingston personnel have helped with STRETCH at one time or another & are now working on it there. About ten such systems are envisioned. It should be stressed, however, that STRETCH is an all DSD program at the moment.

Present DSD plans are to disband the STRETCH (Poughkeepsie) group completely sometime early next year. The personnel will be reassigned to other projects in Poughkeepsie. A number of men have already been transferred -- including several key planners and junior managers.

There seems to be some difference of opinion as to what the feature of STRETCH will be. One group in DSD is planning to institute an engineering redesign of STRETCH using STAM-PAC circuits. The redesign would take place in Two steps: (1) Exchange, High speed exchange & Memory bus, then (2) Arithmetic unit & controls.

← Estimates are that such a redesign could be placed in ~~effect~~ after the 5th production machine -- probably early in 1962. A sizable reduction in manufacturing cost is expected.

Another opinion (held by some ~~responsible~~ DSD managers) is that STRETCH will not be re-engineered, but will be dropped as a "dead-end" machine. DSD would then plan to bring out an altogether new machine for announcement in 1963 which would be equal to or somewhat less than STRETCH performance, but would cost much less. The same people also ~~say~~ that ~~the~~ DSD is losing interest in such large machines & is looking more to the larger market area at the ~~650~~ 650 - 704 performance level.

II. Present Plans for Super-STRETCH

"Super STRETCH" is at present a very vague term which is being applied to a proposal granted by Corporate Research for AEC

Recommendations:

Past history - men of optimism

Present condition
(Finances)
- ACC
- HVST
- FarmBoy

present plans
- distant gaps

manufacture - sales . . .

Q: - pull out of Kingston? - cannot move STR, mfg?

Q: - product improvement of STR, who will do?

Q: what about DSD -- They will go ahead . . . must avoid rivalry of MILCOMP vs [redacted] 7090 Type?

— large machine must be general purpose in a class of [illegible]

present plans to distant groups . . .
better talent -- some advantage but loss of momentum could be great

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Honest delay will reflect on whole company -- FSD will catch brunt on future bus & shifting front

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Possible Plans:

1. STR of HVST stay in FSD [redacted]
Production to FSD [redacted].
no improved machine

2.

FSD Policy

- cannot spend corp. profits on devel.
- MARP money ~~amt~~ small & not for large efforts
- STRETCH improvement plan -- uses Kingston personnel but in DSD program, -- had to get govt money for this --- (sort of thing Corp profits should be spent on)
- question of special modifications
- special I/O etc -- handled as other commercial equip now.

- FSD must consider another 7090 - ~~MILCOMP~~ battle i.e. 2 machines in about same area

could take over whole STRETCH, Prog -- then DSD would buy out another machine.

FSD plans to do a lot of Technology research

FSD can't take over present STRETCH with its financial picture -- if Corporate would give it up -- DSD would immediately come in with cheaper machine

STRETCH has been the inspiration of the world ~~now~~ -- will fade in time

Best Strategy:

- Have DSD turn STRETCH over to FSD minus its huge development bill, but plus the Corporate funds necessary to finish the models -- TRITON is biggest headache..
- Have Kingston Engrg. push the redesign program to get the cost down & the performance up (For this purpose secure a few dozen of the best P'K engrs.)
- ~~Deliver~~ improved version of STRETCH early '62 -- program compat.
- concentrate on I/O applications of interest to govt... Displays - remote inputs -- real time control, etc. - high speed printers.

Strategy (cont)

Re: Super Stretch -- { cannot get big computer contracts any more

- we are in no position to take a machine contract yet -- Technology
- danger - if we don't take machine contract a competitor may ---

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Should be a Research contract -- with FSD keeping on top of situation

Machine contract should be with FSD using Research as subcontractor -- necessary to hold to schedule & goals

- however -- will still need Corporate support for Cyc. program Technology.
